



North South University

Department of Electrical & Computer Engineering

Lab Assessment

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Lab No:	01
Title:	Introduction to ARM Assembly Language
Course Code:	CSE331L.04
Course Name:	Microprocessor Interfacing & Embedded System Lab
Section:	Section 04
Faculty:	Mr. Md Raqibur Rahman [RQN]
Date of Submission:	15 th Feb, 2025
Due Date of Submission:	15 th Feb, 2025
Submitted to:	Md. Wardun Islam Noon

Question:

Write an assembly program using the following instructions:

MOV Move data

MVN Move and negate

ADD Addition

SUB Subtraction

MUL Multiplication

LSL Logical Shift Left

LSR Logical Shift Right

ASR Arithmetic Shift Right

ROR Rotate Right

EOR Bitwise XOR

AND Bitwise AND

ORR Bitwise OR

Answer:

```
.global __start  
__start:
```

```
    @MOV  
    MOV R0, #10;  
    MOV R1, #5;
```

```
    @MVN  
    MVN R5, R0;
```

```
    @ADD  
    ADD R2, R0, R1;
```

```
    @SUB  
    SUB R3, R0, R1;
```

```
    @MUL
```

MUL R4, R0, R1;

@LSL
LSL R6, R0, #1;

@LSR
LSR R7, R0, #1;

@ASR
ASR R8, R0, #1;

@ROR
ROR R9, R0, #1;

@EOR
EOR R10, R0, R1;

@AND
AND R11, R0, R1;

@ORR
ORR R12, R0, R1;